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Fields of Concentration:

Primary Field(s): Labor Economics

Secondary Field(s): Development Economics

Examinations Completed

2022 (Oral): Labor Economics, Development Economics

Dissertation Title: *Essays on Technology, Skills, and Labor-Market Adjustment*

Committee:

Professor Mark Rosenzweig (Chair)

Professor Orazio Attanasio

Professor Barbara Biasi

Education:

Ph.D., Economics, Yale University, 2027 (expected)

M.Phil., Economics, Yale University, 2024

M.A., Economics, Yale University, 2023

B.A., Economics and Mathematics, Yale University, 2018

Fellowships, Honors and Awards:

Cowles Foundation Fellowship, Yale University,

2021 - 2025

Yale Graduate Fellowship, Yale University,

2021 - 2026

Institute for Humane Studies Fellowship,

2023

Kingsley Trust Fellowship,

2026

Research Grants:

2023 Robin E. Evenson Fund, Economic Growth Center, \$2,500 USD

Teaching Experience:

Fall 2023: Teaching Assistant to Prof. Yusuke Narita, ECON 1123, Intermediate Econometrics and Data Analysis, Yale College

Fall 2024: Teaching Assistant to Prof. Benjamin Polak, ECON 2159, Game Theory, Yale College

Spring 2025: Teaching Assistant to Prof. Penny Goldberg, ECON 1115, Introductory Microeconomics, Yale College

Fall 2025: Teaching Assistant to Prof. Benjamin Polak, ECON 2159, Game Theory, Yale College

Research Experience:

Research Assistant, Prof. Orazio Attanasio, Yale University, 2023 - 2024

Research Assistant, Prof. Yusuke Narita, Yale University, 2023

Work Experience:

Research Assistant, International Monetary Fund, Washington, DC, 2019 - 2021

Publications:

Boz et al. (2022). "Patterns of Invoicing Currency in Global Trade: New Evidence." *Journal of International Economics*, 136, 103604.

Working Papers:

"Workforce Demographics and New Software Adoption," with Sabrina Peng, Job Market Paper, November 2026.

Work In Progress:

"Perceived Education Returns Abroad and Investment at Home," with Emma Naslund-Hadley, Siu Yuat Wong, and Qian Yao Ye, November 2026.

Referee Service: *Review of World Economics*

Languages:

Vietnamese (native), English (native), Chinese (fluent)

References:

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Dissertation Abstract

“Workforce Age Structure and Frontier Technology Adoption: Evidence from Newly Released Software”, with Sabrina Peng [Job Market Paper]

There is growing concern that workforce aging may negatively affect economic growth, along with continuing debate over policies to ameliorate labor-force aging. One major determinant of productivity growth is the speed at which new technologies are adopted. A common hypothesis is that older workers may be less inclined or less able to adopt new technologies, but there is little systematic quantitative evidence on this phenomenon. This paper examines the relationship between worker age and technology adoption by estimating the effect of local workforce age composition on firms' adoption of newly released software technologies. Using novel data based on Lightcast job postings and 15 matched legacy and new software technologies released between 2012 and 2016, the analysis quantifies the causal effect of the local share of college-educated workers ages 50 to 65 on firm demand for the newest software technologies. To address the concern that regional age structure may be endogenous, we instrument the local old-worker share using a shift-share measure based on historical commuting-zone age composition and national survival rates, while allowing for a rich set of controls.

The empirical results show that firms are significantly less likely to seek new workers with skills in the newest technologies when they are located in labor markets where younger workers are more scarce and thus command higher wages. Moreover, based on a survey of computer scientists that elicited information on the productivity gains from the new software technologies and their learning challenges, the estimates indicate that worker age has a stronger negative effect on adoption when the new software is more difficult to master. The evidence is weaker that aging is a barrier to adoption for new software that is relatively more productive compared with its legacy counterpart.

The empirical findings are used to construct and calibrate a model of worker and firm behavior. In the model, firms choose between production using an incumbent technology and a more productive frontier technology, while workers decide whether to pay a cost to acquire the skill needed to use the frontier technology. The model incorporates three mechanisms that differentiate the behavior of young and old workers. First, older workers have shorter time horizons than younger workers, given finite working lives, which reduces the benefits they receive from costly investments. Second, older workers have more experience with legacy technologies, so the gains from adopting new technologies are lower for them than for younger workers. Third, older workers may be inherently less able than younger workers to learn new technologies.

The calibrated model permits counterfactuals showing that policies affecting the age composition of the workforce, such as increased immigration of younger workers, increased fertility rates, and altered incentives for retirement, have different effects on technology adoption. In particular, aging caused by weak inflows of young workers, as under low fertility or low migration, can slow adoption by reducing the share of workers with long retraining horizons. By contrast, longer expected working lives, as under delayed retirement, can mitigate this force in the model by increasing older workers' return to skill investment.

“Perceived Education Returns Abroad and Investment at Home,” with Emma Naslund-Hadley, Siu Yuat Wong, and Qian Yao Ye [Work in Progress]

This project studies how expectations about migration shape educational investment in origin countries. In high-migration settings, families decide how much to invest in children's schooling while facing uncertainty about earnings abroad, migration risks, and returns to schooling if children remain at home. The study uses a randomized information experiment with caretakers of children ages 12 to 14 in El Salvador, recruited through public schools and local partners. Participants receive experimentally varied information about U.S. wages for Salvadorans, migration success probabilities, wage uncertainty, and education returns. The design then measures how information changes beliefs, migration intentions, and willingness to invest

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in supplemental education. The project tests a central mechanism in the brain-gain hypothesis: the idea that the possibility of migration can increase schooling investment before migration occurs.